

Informing Jobseekers about Shortage Occupations: Effects on Training Choices*

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Abstract

In a field experiment involving 100,000 recently unemployed individuals, we investigate how jobseekers' training decisions react to information about labor shortages. The intervention increased training participation by 6%, driven by an increased take-up of transversal programs (+10%) rather than specialized occupational training. Further analyses show that the intervention also reorients job search toward shortage occupations. At the same time, the increase in transversal training is not disproportionately concentrated among jobseekers who also shift their search toward shortage occupations. These findings suggest that jobseekers respond to information on labor shortages both by investing in relatively broad skills and by redirecting their occupational search toward jobs in higher demand.

JEL codes: J24, J62, J68.

*We warmly thank Michèle Belot, Koen Declercq, Robert Dur, Anne Gielen, Bart de Koning, Jean-William Laliberté, Lukas Lehner, Pierre-Guillaume Méon, Esther Mbih, Emilie Rademackers, Roland Rathelot, François Rycx, Arne Uhlendorff, Joe Vecchi, and Sunčica Vujić for their valuable suggestions. This work benefited from discussions with participants of the SOLE, AASLE, EALE, ESPE, BDLE, AFE and EEA Annual Conferences, as well as the 7th LISER and IDSC of IZA Workshop: Matching Workers and Jobs Online, ZEW Conference on Research with and within Organizations, Bristol Applied Economics Meetings, DULBEA-ROA Workshop on Information Frictions and Job Search, and Field Days. We are also grateful for helpful comments of seminar participants at Leiden, Maastricht, Turku, Rotterdam, Saint-Louis, Ghent, Antwerp and ULB. We thank Jean-Marc Manfron, Paul-Louis Colon, Philippe Doneux, and Anne-Céline Adnet from the Forem, for their involvement in this project. Benjamine Dejardin and Annanya Chetwani provided excellent research assistance. Elisabeth Leduc gratefully acknowledges the Fonds National de la Recherche Scientifique (FRS-FNRS) for financial support. This study is registered in the AEA RCT Registry under the number AEARCTR-0008313. IRB Approval was granted by the Ethics Committee of the Université libre de Bruxelles.

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1 Introduction

Structural transformations linked to technological change and globalization are fundamentally reshaping what skills employers require from their workers. As a result, many countries are investing heavily in demand-driven training programs aiming to facilitate occupational transitions of unemployed workers toward high-demand occupations. A growing body of evidence suggests that these policies can substantially improve participants' employment prospects (Grosz, 2020; Katz et al., 2022; Humlum et al., 2025), yet training participation remains low in most countries, especially among disadvantaged groups (OECD, 2020). Better understanding how jobseekers make decisions about training is therefore crucial to addressing labor shortages, as well as tackling long-term unemployment.

In this paper, we explore how jobseekers' training choices respond to information about labor market tightness. While employers appear to favor longer specialized training over shorter programs, especially in tighter labor markets (Azmat et al., 2025), enrolling in occupational training requires individuals to invest in new skills and to reconsider their occupational aspirations. In turn, jobseekers may interpret shortage information as revealing skill gaps, but may not view this as a reason to enter occupation-specific training. Some may instead favor broader skill investments that preserve occupational flexibility and better align with their preferences. Consequently, sharing information about labor shortages could lead them to pursue occupation-specific training for jobs in demand, or instead to invest in more transversal skills that may improve their employability across a broader set of sectors and occupations.

We ask whether jobseekers respond to information about labor market tightness by investing in occupation-specific skills for high-demand jobs, or instead by choosing broader, more transferable training. To answer this question, we implement a large-scale information intervention that exposes unemployed individuals to information about shortage occupations and related training opportunities. We then study how jobseekers translate this information in terms of training decisions, and specifically whether they respond by enrolling in occupation-specific programs or by investing in more transversal skills. By doing so, we contribute novel insights on training decisions in the context of skill mismatches between unemployed jobseekers and employers' hiring needs.

To explore this question, we ran a field experiment in partnership with a Public Employment Service (PES henceforth) in Belgium, in which all individuals who registered as unemployed between July 2021 and December 2021 (roughly 100,000 individuals) were sent an invitation

to complete a survey. For a randomly selected half of this population, the email also included information about shortage occupations, the availability of training programs offered, and the conditions under which they could enroll. The intervention primed jobseekers to think about shortage occupations and training before their first caseworker meeting, when they could receive more personalized guidance. By targeting a broad set of recently unemployed jobseekers and covering a wide range of shortage occupations, our intervention aimed to guide jobseekers toward occupations in demand early in their unemployment spell, helping them avoid unnecessary delays at very low cost and without presupposing which specific occupation would best match their interests.

We use administrative records, complemented by data on email engagement and survey responses, to estimate the effects of the intervention on training participation, its relationship with job search behavior around shortage occupations, and, ultimately, employment. We first show that the intervention raised training participation by 0.36 pp 6 months after the intervention took place, a 6.13% increase with respect to the average rate of enrollment in the control group. All effects on training enrollment materialize in the first 6 months after the intervention, with effects remaining at 0.36 pp after 12 months, and are accompanied by increases in training completion of the same magnitude.

We then investigate whether this increased training participation is driven by “transversal training” – defined as courses teaching skills that are applicable to a wide range of sectors and occupations (e.g., language courses) – or “occupational training” – defined as programs that focus on the acquisition of skills that are specific to an occupation or sector. We find that the observed changes in training behavior are essentially driven by an increased participation in transversal training, whereas participation in occupational training and programs targeting specifically shortage occupations remained unchanged. These findings suggest that the intervention increased engagement with training, but that this response did not take the form of direct reskilling into shortage occupations, pointing instead to broader skill development that could facilitate transitions into a range of occupations.

Next, we shed light on the mechanisms that could explain why treated jobseekers enroll in transversal rather than occupational training programs. We first focus on jobseekers’ beliefs and preferences regarding training and employment prospects. Survey evidence we collected for this paper indicates that jobseekers view transversal training as more useful for improving their employment prospects than training for shortage occupations. Since finding a job is their primary motivation for enrolling in training, and the intervention affected beliefs about both types of programs similarly, this perception likely led jobseekers to favor transversal

over occupational training.

Second, we examine whether supply-side features of the PES training offer can explain why the intervention increased participation in transversal rather than occupational training. Using administrative records and PES survey data, we compare transversal and occupational programs in terms of duration, frequency, variety, and perceived quality. We find little evidence that transversal trainings were easier to access or more attractive on the supply side: they are not shorter, not offered more frequently, do not provide a wider range of options, and are rated slightly lower by participants. This suggests that the stronger response in transversal training is unlikely to be driven by the training offer itself, reinforcing the role of jobseekers' preferences in shaping their choices.

Third, we examine whether the intervention affected jobseekers' occupational preferences and how these changes relate to training participation. Treatment increased the likelihood of listing at least one shortage occupation among preferred jobs by 1.14 pp (1.86%), reflecting a reorientation of job search toward these occupations. Treated jobseekers became more likely to search for a shortage occupation without participating in transversal training, and less likely to neither search for a shortage occupation, nor participate in transversal training. At the same time, although the intervention increases the probability of jointly participating in transversal training and searching for a shortage occupation, this effect is not significantly larger than the corresponding effect without searching for a shortage occupation. The absence of an effect on occupational training therefore does not appear to reflect a failure to process the information about shortage occupations. Jobseekers clearly incorporate this information into their search choices, while the disconnect arises at the human-capital investment margin, where they appear willing to consider different occupations but not to commit to the specialized training required to enter them.

Ultimately, changes in training and job search behavior do not translate into detectable increases in employment probability. Examining the magnitude of the effects suggests that the impacts on training and job search were too modest to potentially translate into meaningful employment gains. However, given the low marginal cost of the intervention and the potentially high rewards from training (Humlum et al., 2025), informing jobseekers about shortage occupations and available training programs could still be worthwhile.

Together, these findings make a substantial advance on the literature by documenting how jobseekers' decisions around training investments react to labor market information. They speak most directly to the literature on job search strategies to enter high-demand occupations. Existing evidence shows that jobseekers are willing to redirect their search toward

high-demand occupations, which can increase employment (Belot et al., 2019; Belot et al., 2026; Altmann et al., 2026), especially when the recommended occupations align with job-seekers’ existing skills (Bächli et al., 2025). When transitions into shortage occupations require reskilling, however, it remains unclear how jobseekers respond to information that combines labor-market demand with available training opportunities.

This question is particularly relevant given the existing evidence on active labor market policies that not all training programs are equally effective. Positive effects of training tend to be stronger in the case of vocational programs than for transversal programs (Stenberg and Westerlund, 2015; Wood et al., 2024), especially if the training is demand-driven and tailored to local employers’ needs (Grosz, 2020; Baird et al., 2022; Katz et al., 2022; Foged et al., 2026; Andersson et al., 2024; Anwar et al., 2024). Yet, despite these potential returns, training take-up remains low (Eurostat, 2023). Our findings show that a low-intensity intervention combining information on training opportunities and shortage occupations can significantly increase training participation, while also shifting job search toward shortage occupations.

By doing so, we also contribute to a broader literature studying how information frictions shape the effectiveness of public policies (Chetty and Saez, 2013; Hoxby and Turner, 2015; De Neve et al., 2021). Recent papers have focused specifically on information interventions supporting better job search strategies, encouraging jobseekers to broaden the set of occupations they consider, or offering practical tools to strengthen their applications (Altmann et al., 2018; Briscese et al., 2021; Wheeler et al., 2022; Le Barbanchon et al., 2023; Behaghel et al., 2025; Altmann et al., 2026; Klæui, 2026). Relatedly, some recent papers study information frictions in training decisions. They show that correcting jobseekers’ beliefs about labor market prospects can improve selection into training and increase willingness to enroll, while information on training costs and returns has mixed effects on enrollment (Barr and Turner, 2018; Dhia and Mbih, 2023; Lergetporer et al., 2023; Chakravorty et al., 2024; Lehner and Schwarz, 2024; Bhatiya et al., 2026). We contribute to this literature by showing that information can not only redirect job search toward higher-demand occupations, but also encourage the investment in training that may be necessary when jobseekers do not yet possess the skills required for these occupations.

Finally, our findings speak to an emerging literature exploring human capital investment decisions of workers and jobseekers. Recent work suggests that occupational preferences play an important role in training decisions (Dejardin et al., 2026; Delfino et al., 2026), as do personality traits such as risk preferences (Caliendo et al., 2023) and locus of control (Caliendo et al., 2022). From the firm’s perspective, evidence suggests that employers op-

erating in tight labor markets have a high demand for workers having followed specialized training (Azmat et al., 2025), but are reluctant to offer on-the-job training themselves because of poaching concerns (Adams et al., 2026). Yet little is known about how jobseekers incorporate information about labor market tightness in their training strategy.

Our findings show that this link is more complex than previously understood. Although the intervention increased training participation, this effect was driven mainly by transversal rather than occupational training. This suggests that information about labor shortages does not automatically lead jobseekers to invest in occupation-specific skills, even when such programs may offer high returns. One mechanism we emphasize is that jobseekers expect higher returns from transversal than occupational training. This suggests training for shortage occupations may be perceived as a narrower, and therefore riskier, investment than transversal training, which provides more broadly transferable skills. This helps explain why take-up of demand-driven training programs can remain low, even as the intervention shifts jobseekers' search toward shortage occupations.

2 Institutional Setting

The PES with which we partnered for this evaluation is responsible for all active labor market programs in the region of Wallonia, in Belgium.¹ Wallonia has a relatively large pool of unemployed jobseekers; there are on average approximately 200,000 unemployed jobseekers at any point in time (12.6% of the active population), 120,000 of whom have been unemployed for more than a year. At the same time, many employers in Wallonia struggle to fill the vacancies they open; in 2022, the vacancy rate was 3% on average (Statbel, 2025).

The PES's key mission is, on the one hand, to guide jobseekers and workers in their careers and, on the other hand, to meet companies' recruitment and skills needs (De Brouwer et al., 2023). In this context, one of their key activities is to offer and promote training programs, which can broadly be classified into two types: transversal and occupational trainings. Occupational (or vocational) training courses prepare participants for a specific job, profession or sector. In contrast, transversal trainings offer a more general curriculum that covers a wider range of subjects and skills. Figure 1 illustrates, for each training type, the ten programs occurring the most frequently in the data.

¹Belgium is particularly well-suited to study the question of training strategies to enter shortage occupations because it has a higher unemployment rate than the OECD average and, at the same time, a larger number of unfilled vacancies. Therefore, although Belgium's vacancy-to-unemployed ratio is around the OECD average – making it representative of developed countries –, unemployment and shortages affect a relatively larger portion of its labor market.

Although the PES and its partners offer a wide range of transversal and occupational training programs, they tend to put the most emphasis on vocational training for which firms operating in the region face skill shortages (referred to as “trainings for shortage occupations” throughout this paper). These shortages can be of a qualitative nature (e.g., candidates who apply for the jobs do not have the qualifications that employers require) or of a quantitative nature (i.e., there are too few unemployed jobseekers to fill the job openings). Each year, the PES analyzes vacancy statistics and consults employers to make a list of jobs that are in shortage. It then attempts to direct particular efforts to alleviate these shortages, primarily through training the unemployed.²

But despite the large number of training options offered, shortage occupations typically remain understaffed for many years, and trainings tend to be chronically under-subscribed.³ This begs the question of why jobseekers do not decide to enroll, especially given the high potential rewards and the attractive conditions under which these programs can be pursued. At the time of the experiment, individuals enrolled in training indeed remained eligible for unemployment benefits, and were in some cases exempted from their job search requirements.⁴ They were also eligible for a travel allowance, extra childcare benefits, and special interview coaching. Those enrolled in a training for a shortage occupation were additionally entitled to a financial bonus ranging from €350 to €2,000 if they successfully completed the program. Moreover, getting a job at the end of the training is very likely given that the occupation is, by definition, understaffed. In spite of all this, many jobseekers remain unemployed for long periods of time without ever participating in training.

We examine whether reducing information frictions about shortage occupations encourages jobseekers to enter training, which types of training they choose, and why. Although the PES’s website provides some information on shortage occupations and training opportunities, one needs to actively search the website to find this information. In theory, caseworkers could also provide job search advice and information about training opportunities, but they are not instructed to systematically focus on shortage occupations or trainings during their meetings with jobseekers. As a result, jobseekers in Wallonia are unlikely to be informed

²Appendix A.1 describes in more detail how the PES identifies shortage occupations and shows the full list of occupations that were considered in shortage in 2021 and 2022. In 2021 (2022), out of 650 occupations, 121 (126)—or almost 20%—were defined as being in shortage in Wallonia. The jobs in this list are quite diverse (e.g., healthcare sector, construction, service industry, logistics, industry) and require varying levels of education. The length of training required to enter these jobs ranges from a few weeks to a few years. Overall, most unemployed individuals considering upskilling or reskilling should be able to find a shortage occupation that matches their education level and preferred sector of work.

³On average, between 2022 and 2024, only about 53% of all available training spots offered by the PES and its partners were filled. Anecdotally, many Employment Agencies seem to face a similar issue.

⁴This is true under a few broad conditions listed in Appendix A.2

about shortage occupations and related training programs unless they first express interest in one. In this context, an information intervention has the potential to encourage more jobseekers to enter shortage occupations and, if they currently lack the skills to do so, to enroll in the appropriate training programs.

3 Experimental Design

3.1 Intervention

We partnered with the Walloon PES and implemented a large-scale field experiment that took place between October 2021 and February 2022. The intervention consisted in sending information, by email, about shortage occupations and trainings opportunities. All 98,847 jobseekers who registered at the PES between July and December 2021 and had an email address were included in the experiment.⁵

The PES sent individuals an invitation to complete a satisfaction survey regarding the services they had received while unemployed.⁶ The email sent to the control group solely included the invitation to complete the satisfaction survey. For a randomly selected half of individuals who were assigned to the treatment group, the email also contained information about shortage occupations and related training opportunities.

The treatment email informed jobseekers that shortage occupations systematically lacked candidates, and that focusing on these occupations could allow them to increase their probability of (re-)entering the labor market. The number of shortage occupations (126) was mentioned in order to signal that there were many options to choose from, without needing to add the full list to the email. The message also mentioned that jobseekers could follow trainings, free of charge, and under relatively attractive conditions. In particular, it stated that they maintained rights to their unemployment benefits without having to actively search for a job during the training, that they could get additional childcare and travel allowances, and that they were entitled to a bonus ranging from €350 to €2,000 after completing a training for a shortage occupation. Interested jobseekers were then referred to their local

⁵In the first wave of the experiment, the email was sent to 60,485 individuals on October 19, 2021 (Tuesday around 9am). A reminder was sent on October 26 (Tuesday around 2pm) to those who had not opened the first email. In the second wave, the email was sent to 38,362 individuals on February 1, 2022 (Tuesday around 9am). A reminder was sent on February 8, 2022 (Tuesday around 2pm) to those who had not opened the first email. According to the PES, this time of week and day maximizes the rate of engagement with the emails they send to jobseekers.

⁶This satisfaction survey is usually sent to a small sample of recent jobseekers but, for the purpose of our experiment, the PES sent the survey to all recently registered jobseekers.

“Customer Service” – agencies that are in charge of assisting jobseekers at different stages of their unemployment spell – whom they could contact for additional information. At the end of the email, there were three links leading to additional information on shortage occupations, namely: (i) a video explaining what shortage occupations are, (ii) the full list of shortage occupations, and (iii) additional information on the financial bonus. A copy of the treatment and control emails are presented in Appendix A.3.

The treatment was thus intended to steer jobseekers toward more promising occupations and related training opportunities. It did so by informing them, in a salient and simple manner, about the existence of shortage occupations and the attractive conditions under which they could seek training while unemployed. The emails were sent before the first meeting with a caseworker would have taken place.⁷ This means that, even if jobseekers did not take immediate action by themselves after receiving the email, they had an opportunity to steer the conversation with their caseworkers toward the shortage occupation of their choice and could then obtain more tailored information (about job search or training) from their caseworker directly. In this sense, the timing of the intervention and the choice of the target population allowed to maximize the potential effectiveness of our low-cost, low-intensity, information intervention.

To summarize, we implemented a low-cost information intervention, providing general information about shortage occupations and training opportunities to a broad population of recently unemployed jobseekers. In turn, the information provided aimed at encouraging jobseekers to seek additional information with their local Customer Service or during their upcoming meeting with a caseworker. The intervention was intentionally broad on both sides of the market – both in terms of the jobseekers it targeted and the shortage occupations it highlighted – leaving ample room for individuals to determine, with the support of a PES professional, which specific course of action would be most beneficial for them. As described in the next section, our data allows us to follow jobseekers for up to 14 months after the intervention took place. This leaves ample room to observe effects that would occur in a sequential way, i.e., after the jobseekers meet their caseworker.

3.2 Randomization

Half of the individuals in our sample were randomly selected to receive the treatment email. Before sending the treatment and control emails, the PES sent us the anonymized identifiers

⁷At the time of the experiment, these meetings typically took place about four months after registering at the PES, and individuals in our experiment had been unemployed for less than four months at the time the emails were sent.

of all individuals who had recently registered. We then performed the randomization based on the (meaningless) last digit of the anonymized individual identifier.

Table 1 shows descriptive statistics on the characteristics of the treated and control groups, before the emails were sent. Reassuringly, the characteristics of the two groups are well-balanced. None of the differences between the treatment and control groups is statistically significant, except that there are slightly (0.1 pp) more individuals who live outside the Walloon region in the control group. This difference is very small, and only marginally significant; it therefore does not threaten the validity of our findings.

3.3 Engagement with the Email and its Content

The advantage of using emails to communicate our treatment information is that we can reach a large group of jobseekers at a very low cost. In terms of data, it allows us to observe who opens the email, and thus who could actually be affected by our intervention. The data on email engagement is linked to administrative data, which means we can also assess who engages with the email and its content (i.e., who clicks on the provided links).

Table 2 shows information on email engagement for the overall sample (column 1), the treatment group (column 2), the control group (column 3) and the difference between the treatment and control groups (column 4). Approximately 54% of individuals opened the email they were sent by the PES. This share is the same in the treatment and control groups, which is reassuring as it indicates that there was no differential selection into reading the email between the two groups.

Table 2 further shows that treated individuals were more likely to click on at least one link in the email (13.92% versus 11.31% of the controls) and clicked on more links on average, which reflects the fact there were more links they could potentially click on. Interestingly, the treatment group was less likely to click on the survey link (7.81% against 11.08%), suggesting that the information on shortage occupations and related training programs diverted jobseekers' attention away from the survey. Similarly, treated jobseekers were slightly less likely to click on the social network links of the PES (0.19% against 0.33%).

Table A1 in the Appendix shows how the treatment group interacted with the information on shortage occupations and related trainings. It shows that 7.65% of treated individuals who opened the email clicked on at least one of the treatment links, indicating a certain interest in the information they were being provided. Treated jobseekers were most interested in the list of occupations in shortage, with 4.66% of them clicking on that link. Additionally, 1.65%

of them clicked on the video presenting the occupations in shortage, 1.47% clicked on the link referring them to the Customer Service and, maybe surprisingly, only 1.24% clicked on the link that provided them with more information on the financial incentives.

4 Empirical Approach

4.1 Data

In our main analyses, we rely on administrative data containing information on personal characteristics, training participation, employment spells, and job search. This data contains highly reliable and extensive information on all individuals who were part of the experiment. The data on personal characteristics contains information on age, gender, nationality, education level, field of education, district of residence, and the existence of previous unemployment spells. The training data includes all training activities undertaken by jobseekers, with the name of the training activity, its start and end dates, and the reason for ending the training. The employment data contains information on the start and end date of each spell, as well as the sector of activity in which the employer is active. Finally, the job search data contains a list of occupations for which each individual is listed as searching for a job in, as well as the number of times they have connected themselves to their personal space on the PES website. We use these datasets to construct our outcomes of interest, pertaining to training behavior, job search, and employment participation.

First, to investigate whether the information intervention affected training behavior, we consider whether individuals participate in any training activity 6 (short-term estimate) and 12 (long-term estimate) months after the intervention. To this end, we create a binary variable equal to one if the individual is observed taking part in at least one training activity, and zero if not. Since we are interested in understanding how jobseekers respond to information about shortage occupations, both in terms of training participation and training choice, we subdivide training activities into transversal and occupational programs. Transversal training courses include training activities that do not directly target a given occupation or sector, such as a language course, trainings for using IT tools, or orientation and coaching. In contrast, occupational training programs focus on acquiring skills specific to an occupation or sector. These categories are mutually exclusive and exhaustive, i.e., we classify all training activities as either transversal or occupational. In a second step, we subdivide occupational trainings between those that target occupations that are in shortage and trainings that target occupations that are not in shortage.

Second, we exploit data from the satisfaction survey to shed light on the mechanisms that may explain treated jobseekers’ participation in training and their choice of program.⁸ This data includes respondents’ answers to questions relating to their beliefs about job search strategies and training intentions. Specifically, we asked respondents in the treatment and control group: (i) whether or not they had the intention to follow a training in the coming year, and (ii) how useful they considered different job search strategies to be. For intentions, we build a dummy variable equal to one when survey respondents answered that they had the intention to follow a training in the coming year. For beliefs about the usefulness of job search strategies, we build a dummy variable equal to one if the respondent considered a strategy as being “useful” or “very useful”. We asked about the following job search strategies: searching for a job in a shortage occupation, participating in training programs, participating in trainings for shortage occupations, having a good CV and motivation letter, using one’s network, meeting one’s caseworker, making use of employer subsidies, or searching for a job in which they already have experience.⁹ More details on the survey and the questions it contains can be found in Appendix A.4.1.

To further investigate the underlying mechanisms, we also examine whether the intervention affected jobseekers’ occupational preferences and how such changes are related to subsequent training participation. To do so, we define three job search-related outcomes. First, we construct a dummy equal to one if an individual lists a shortage occupation among their list of occupational preferences.¹⁰ We also consider the total number of occupations that jobseekers list in their occupational preferences. Moreover, we observe the number of times individuals connect to their personal space, which we use as a proxy for job search intensity.¹¹

Finally, we are interested in estimating the effect of the intervention on the probability of finding a job. To this end, we use data on all employment spells that individuals experience over our period of analysis and construct a dummy equal to one if they experience any employment spell starting in the first 6 months (for short-term estimates) after the intervention took place. Long-term estimates consider all employment spells that start within the first 14 months after the intervention.¹²

⁸The survey and administrative data cannot be linked to one another.

⁹Possible responses ranged from: “no impact” to “very high impact” on a five-point scale.

¹⁰Jobseekers must list the occupations in which they would like to find employment in their personal space on the PES website. We can observe the list of these occupations 6 and 14 months after the intervention. Jobseekers can list as many occupations as they like, but typically have between one and six.

¹¹This variable contains the total number of connections 6 and 14 months after the intervention.

¹²In Appendix A.5, we check that our findings remain stable when we consider only employment spells lasting at least five consecutive days, or when looking at unemployment exit.

4.2 Regression Specification

Because individuals were randomly assigned to treatment, we can estimate the following equation to recover intention-to-treat (ITT) effects:

$$Y_i = \alpha_1 + \beta_1 Treatment_i + \epsilon_i \quad (1)$$

A key advantage of delivering the information treatment by email is that we can observe who opens the email, and thus who actually accesses the information. Therefore, throughout our main analyses (i.e., those using administrative data), we restrict our sample to the population who opened the email and were thus effectively part of the experiment ($N = 53,822$). This means our ITT estimates measure the effects of the intervention for the group of individuals who would be reached by this email, if it was adopted by the PES as a standard policy for all individuals registering as unemployed.¹³ The robustness section explores how our estimates vary depending on the criteria used for inclusion in the sample.

We estimate treatment effects without controlling for personal characteristics, as these are balanced between the treatment and control groups. We nevertheless show in Section 6 that our findings are unaffected by their inclusion. In those analyses, control variables include age, gender, nationality, education level, province of residence and a dummy indicating whether an individual has experienced unemployment spells before.

5 Findings

5.1 Effects on Training Participation

5.1.1 Benchmark Effects

We first examine whether the intervention affected training behavior. Column (1) of Table 3 shows the effect of the intervention on the probability of enrolling in any training program. We find that receiving the treatment email increased the probability of enrolling in a training program by 0.36 pp, or 6.13% relative to the control mean. This effect subsists 12 months after treatment, although it becomes marginally insignificant (p-value=0.1107). All of the

¹³We also estimated a local-average-treatment-effect (LATE) for individuals who accessed the information on shortage occupations and related trainings, through clicking on at least one of the treatment links in the treatment email. In these exercises, we instrumented a dummy variable for accessing the treatment information (i.e., clicking on at least one of the treatment links) by the treatment assignment indicator. We then estimated the effect of actually accessing the treatment information in the links on our outcomes of interest, in a two-stage-least-squares approach. These LATE estimates are consistent with our benchmark findings but, because the exclusion restriction likely does not hold, we do not discuss these findings further.

effect on training enrollment materializes in the first 6 months after the intervention, as illustrated by the identical point estimates 6 and 12 months post-intervention. In Appendix A.5, we further examine the effects of the intervention on training completion and duration. We find that the increase in training enrollment is accompanied by an increase in training completion of the same magnitude. This shows that the individuals who were induced to enroll in training as a result of the intervention completed the full training program. Moreover, we find that, although treated jobseekers do not spend more days enrolled in training 6 or 12 months after the intervention, they are more likely to participate in relatively longer training activities (i.e., those lasting more than five or 30 days). This suggests that the intervention may have shifted participation away from several shorter training activities toward fewer but longer (and potentially more “substantial”) programs.¹⁴

Next, we explore whether the information treatment influenced the type of training chosen by participants, distinguishing between transversal programs and those oriented toward specific occupations. In columns (2) and (3), we first subdivide training activities from column (1) into trainings that offer participants the opportunity to learn general skills that can be used in many sectors and occupations (“Transversal Training”) versus programs that teach participants a specific occupation, or at least are specific to an industry (“Occupational Training”). We find that the increased training participation found in column (1) is exclusively driven by increased enrollment in transversal training courses, whereas there is no effect on occupational training programs. This is true both in the short- and long-term.

The absence of an effect on occupational training could nevertheless hide other types of behavioral changes. Specifically, participants who wanted to pursue occupational training might have shifted from their initial occupational preference to a shortage occupation after receiving the information treatment. In this case, we would observe no change in overall participation in occupational training but an increase in enrollments for shortage occupations. Such a shift would indicate that the information intervention effectively reoriented jobseekers toward higher-demand jobs through training. Therefore, in columns (4) and (5), we differentiate occupational training programs targeting shortage occupations from those focused on other (“non-shortage”) occupations. Our findings do not suggest any shift from non-shortage to shortage occupational training programs.

All in all, our findings suggest that even a low-touch information intervention about oppor-

¹⁴When regressing the number of training activities on treatment, we indeed find negative (but statistically insignificant) coefficients. Moreover, Table A3 in the Appendix shows that the intervention affected only participation in actual training programs, while it did not affect participation in preparatory programs (or “pre-trainings”).

tunities in shortage occupations can increase enrollment in training programs. This finding seems to be broadly in line with the literature on the effect of light-touch information interventions in the context of job search (e.g., Altmann et al., 2018; Belot et al., 2019; Wheeler et al., 2022). In the limited literature focusing on information interventions in the context of education decisions for the unemployed, results are more mixed. For example, Dhia and Mbih (2023) observe no effect on training participation of an intervention that highlighted the returns and costs of specific trainings, as well as the simplicity of registration procedures. A possible explanation for the absence of an effect in their case is that the intervention targeted a specific subset of trainings and jobseekers; the emails focused on specific trainings and were sent only to jobseekers already registered as interested in those occupations, who may not have needed the information. In contrast, Barr and Turner (2018) show that after 6 months, a general letter encouraging unemployment insurance recipients to enroll in a post-secondary program increased training take-up by 40%. In their case, the intervention included an “assistance dimension” (i.e., unemployment insurance recipients were also sent information that facilitated engagement with a number of intermediaries to assist individuals with enrolling), which may have contributed to these larger effects.

Ultimately, our results show that, while a low-cost information intervention can increase training participation, the behavioral response does not correspond to direct skilling into shortage occupations. Instead, the intervention has encouraged investment in more general skill development, potentially facilitating transitions across a broader set of occupations. Section 5.2 investigates why the intervention may have increased enrollment in transversal training programs rather than in training programs targeting shortage occupations.

5.1.2 Heterogeneous Effects

This section examines whether the intervention had stronger effects on training participation for some groups of jobseekers than for others. Although our limited statistical power prevents us from detecting significant differences across groups, exploring which individuals appear to drive the results remains informative. We first explore potential (pre-registered) heterogeneity along demographic characteristics, such as gender and education.¹⁵ Figure 3 shows that all of our effects on training participation are driven by women. This is consistent with Barr and Turner (2018) who also found that the Pell Letter had larger effects on training enrollment among women. A priori, it is unclear why women would be more affected by the information intervention. However, it is notable that women were also more

¹⁵Appendix A.6 also includes a heterogeneity analysis according to age and nationality. It shows that young and older individuals were affected similarly, but that training participation increased more among Belgians than non-Belgians.

likely to open the email and to complete the survey, suggesting that they might have been more attentive to the email sent by the PES. Given that women are overall less likely to participate in training than men, the intervention allowed to diversify somewhat the set of jobseekers enrolling in training along this dimension.

Next, Figure 3 shows that the effect of the information intervention on the probability of participating in training programs is driven by jobseekers with relatively lower levels of education (i.e., those with a secondary school degree or less). The fact that lower-educated jobseekers react more strongly in terms of training participation raises the question of whether jobseekers with different (expected) benefits from training reacted differently. In order to explore this, we perform a heterogeneity analysis where we divide our sample between those with a relatively high employment (respectively training) probability, versus those with a low probability of employment (respectively training).¹⁶

The results are shown in Figure 4.¹⁷ They suggest that jobseekers who are induced to participate in training through our information intervention have a relatively high ex-ante training probability, but a low ex-ante employment probability. In other words, the information intervention fostered training participation among individuals who were on the margin of participating, but were unlikely to find a job. The fact that training enrollment increased among those with a low employment probability is reassuring, as it indicates that the additional training participants are unlikely to have found a job in the counterfactual, and therefore that the intervention did not divert individuals from employment to training.

5.2 Exploring Mechanisms

We examine three mechanisms that may explain why treated jobseekers enrolled in transversal rather than occupational training programs. We first consider the role of jobseekers' beliefs and preferences regarding training and employment prospects. We then assess whether features of the PES training offer can account for the stronger response in transversal training. Finally, we investigate the intervention's effects on occupational preferences, in order to explore if increased transversal training participation was part of a change in job search strategy toward shortage occupations.

¹⁶To do so, we use data from the control group to predict, using pre-treatment characteristics, the probability that each individual enters training or employment in the 6 months that follow the dispatch of the email. We then divide the sample between those with a relatively low (below median) and high (above median) probability of entering training or employment.

¹⁷Appendix A.6 shows additional heterogeneity analyses by unemployment status, benefit receipt, previous UI receipt, and job search focus in $t = 0$

5.2.1 Jobseekers’ Beliefs and Preferences

We use responses from the satisfaction survey sent to all jobseekers included in our experiment to investigate the role of factors related to jobseekers’ preferences and beliefs about training.¹⁸ Overall, these analyses suggest that demand-side factors could explain why the intervention increased training participation, with stronger effects on transversal programs than on specialized training for (shortage) occupations.

We start by analyzing jobseekers’ beliefs about the usefulness of training.¹⁹ Specifically, we first ask jobseekers how useful they consider training to be as a job search strategy, without specifying any type of training in particular. We find that 60.48% of respondents believe that following trainings has a “high” or “very high” impact on their likelihood of finding a job. Of the eight job search strategies we asked them about (see Section 4.1), following a training ranked as the second most useful strategy, right behind searching for a job in which the individual already has experience. This is consistent with recent evidence showing that workers also perceive significant returns to firm training, and that these beliefs predict training intentions and participation (Brosch et al., 2025). In contrast, when asked about the usefulness of following a training for a shortage occupation specifically, only 48.78% of our respondents stated they considered this to have a “high” or “very high” impact on their likelihood of finding a job. In other words, our survey reveals that jobseekers’ perceived returns to pursuing transversal training may be higher than those for training for shortage occupations.²⁰

Second, we examine whether our intervention modified jobseekers’ beliefs about the usefulness of training, and training for shortage occupations more specifically. To do so, in Figure 2, we show the results of estimating Equation 1 with training intentions and beliefs about the usefulness of different job search strategies as the outcomes of interest. Estimates suggest that the intervention increased perceived returns to training in general at least as much as it increased perceived returns to training for shortage occupations. Treated individuals are indeed 3.77 pp (6.4%) more likely to deem training as an effective job search strategy than

¹⁸The survey contains 3,049 responses in total. As in all surveys, the set of respondents is a non-random selection of the overall population and self-selection into the survey is described in Appendix A.4.2.

¹⁹It seems natural to assume that beliefs about the usefulness of trainings would matter for training decisions. Anecdotally, in a more recent survey that we ran for another project and with a different set of respondents, over 35% of jobseekers mentioned wanting to find a job when asked in an open-ended question about their main motivation to enroll in a training program. This was by far the most common response.

²⁰Interestingly, surveys of training participants suggest that employment rates were about 50% higher for occupational training than for transversal programs on average between 2021 and 2025. Although these figures are descriptive and selected, they do suggest that jobseekers’ beliefs about the relative usefulness of transversal versus occupational training may be biased.

respondents from the control group, while this effect is 2.76 pp (5.8%, marginally statistically insignificant) when focusing on training programs for shortage occupations specifically.²¹ At the very least, these estimates indicate that the intervention would not have reduced the gap in perceived usefulness of trainings in general versus trainings for shortage occupations discussed in the previous paragraph.

Taken together, these findings suggest that jobseekers perceive transversal training as more useful for improving their employment prospects than training for shortage occupations. Since finding a job is their primary motivation for enrolling in training, and the intervention affected beliefs about both types of programs similarly, this perception likely explains why the intervention increased participation in transversal rather than occupational training programs.

Other jobseeker preferences and beliefs may also play a role, but our data do not allow us to test these mechanisms directly. One such example is the difference in the perceived riskiness of occupational programs relative to transversal trainings; because transversal trainings teach broadly applicable skills, jobseekers may indeed view them as a safer investment of their time than training for a specific occupation. This interpretation would be in line with the findings of Caliendo et al. (2023), who show that workers' risk preferences matter for the decision to invest in training. Another example is individuals' internal (as opposed to external) locus of control, which has been shown to predict take-up of general (but not specialized) training among German workers (Caliendo et al., 2022).

5.2.2 The PES Training Offer

We investigate to what extent factors related to the training supply can explain why treated jobseekers increase their participation in transversal rather than occupational training programs, including those for shortage occupations. Specifically, we examine differences in training characteristics such as program duration, frequency of starting dates, quality, or choice variety. To do so, we use administrative records on the universe of training spells experienced by individuals in our sample between January 2019 and December 2022, as well as survey data of training participants collected by the PES between 2021 and 2025. Overall, these analyses provide suggestive evidence that supply-side factors play at most a minor role

²¹These estimates grow in magnitude (and significance in the case of trainings for shortage occupations) once we add control variables. However, it should be mentioned that, in Section 4.1, we show that more control individuals clicked on the survey link than among the treated group, suggesting potential differential selection in the survey. Although we observe no differential selection based on observables, we cannot rule out differential selection based on unobservables. Our findings on training intentions and beliefs should therefore be interpreted in light of this potential selection.

in explaining why treated jobseekers decide to enroll in transversal programs rather than in occupational training.

First, jobseekers may prefer transversal training if such programs are shorter than occupational training and therefore require a smaller time investment. In Appendix A.7, we plot the distribution of training duration for transversal and occupational programs, as well as for those targeting shortage occupations specifically. Figure A6 shows that transversal trainings actually tend to be slightly longer than occupational programs; on average, transversal programs last 71 days, compared with 66 days for occupational training and 69 days for trainings in shortage occupations. The distribution of training duration is also flatter for transversal programs, meaning there are more medium-length options. In contrast, occupational trainings offer a larger number of relatively short options. Overall, this points to transversal trainings not representing a lower time investment than occupational training programs.

Second, it is possible that transversal and occupational trainings follow different calendars and that our intervention happened to coincide with a period when more transversal programs were being organized. In Appendix A.7, we plot the number of distinct trainings starting each month between October 2021 (the month the first wave of emails were sent) and July 2022 (6 months after the last information email was sent). Figure A7 shows that, in fact, more occupational trainings start each month than transversal ones. This evidence thus suggests that our results cannot be explained by a larger or more frequent supply of transversal trainings by the PES.

A third potential explanation that is related to training supply, and that could explain why treated jobseekers would enroll in transversal rather than occupational trainings, is the variety of offered programs. Although there are fewer transversal than occupational trainings offered by the PES, the diversity of options may be greater, potentially catering to a wider set of jobseekers. To explore this, we cluster training activities into groups of closely related topics using an embedding model, and assess whether there is a difference in the “variety” of options jobseekers can choose from.²² We find that transversal trainings can broadly be grouped into six different topics, while occupational programs are grouped into 25 topics using the same method.²³ On the one hand, this suggests that occupational trainings offer

²²More specifically, each training option was encoded into a high-dimensional embedding using a “BERT” model, which allows semantically similar trainings to be identified through vector-based indexing.

²³Transversal trainings are grouped into the following topics: (i) tools for job search, (ii) core languages (French, Dutch, English and German), (iii) employability, (iv) creative thinking, (v) basic digital skills, and (vi) specialized languages and communication (e.g., Russian, Mandarin, corporate communication). For occupational trainings, the algorithm groups similar types of occupations (such as butcher, cook, baker and

a wider variety of options than transversal programs. On the other hand, if jobseekers are only interested in one training topic within each training type (e.g., a language training or a training in the food industry), any given transversal training will fit their preferences with a 1/6 probability whereas, for occupational trainings, this will be true for only one in 25 programs.

Finally, we assess whether transversal programs are of higher quality than occupational ones by comparing satisfaction ratings among training participants. For this purpose, we use data from the PES’s annual satisfaction survey of training participants, which reports average satisfaction scores by training type. On average, participants rate occupational trainings at 8.1/10, compared to 7.7/10 for transversal trainings. These figures suggest that differences in training quality are unlikely to explain why treated jobseekers chose to enroll more frequently in transversal rather than occupational programs following the intervention.

5.2.3 Training Choices and Job Search Behavior

To further investigate why information on opportunities in shortage occupations increased training participation, we examine how the rise in training is linked to a reorientation of jobseekers’ strategies toward these occupations.

Column (1) of Table 4 shows that the information intervention increased the likelihood that jobseekers would list a shortage occupation among their occupational preferences. The effect is equal to 0.99 pp 6 months after the intervention, which represents a 1.68% increase. The effect grows somewhat over time, reaching 1.14 pp (1.86%) after 14 months.²⁴ This effect on job search could come from jobseekers adding a shortage occupation to their list, or them changing one or more of their occupational preferences from a non-shortage to a shortage occupation. Column (2) indicates that jobseekers do not necessarily list a larger number of occupational preferences; rather, when they make a change to their list (modification or addition), they are more likely to list an occupation that is in high-demand by employers.²⁵

waiter) together into one topic.

²⁴Interestingly, Figure 2 suggests that the intervention did not affect perceptions about the usefulness of searching for a job in a shortage occupation. One explanation could be that, while jobseekers are aware of the existence of shortage occupations, our intervention made these occupations more salient and, as a result, jobseekers became more likely to list a shortage occupation when they made changes to their list of occupational preferences. In that sense, the information on the existence of shortage occupations would have acted as a reminder rather than a pure information shock.

²⁵Column (3) of Table 4 also shows that receiving the information led to a 0.13 decrease (-7.57%) in the number of connections of jobseekers to their personal space on the PES website. This suggests that treated individuals exerted less job search effort, possibly due to the increased likelihood of being enrolled in a training program.

Overall, these changes in occupational preferences suggest that the increase in training participation may reflect jobseekers’ desire to broaden the scope of their search toward shortage occupations. This result is consistent with evidence from other information interventions showing that encouraging jobseekers to search for high-potential occupations can broaden the scope of their search (Belot et al., 2019; Belot et al., 2026). Our effects are smaller, likely because the information we provide is more general and not tailored to jobseekers’ individual situations and preferences. Differences in the timing and mode of delivery may also matter. In Belot et al. (2019) and Belot et al. (2026), jobseekers receive advice within the job search platform, whereas our intervention provided information by email.

To further understand why the intervention increased participation in transversal rather than occupational training, we examine whether changes in training behavior coincide with changes in occupational search. We estimate these effects using the same specification as in the benchmark analysis, following Equation (1). Table 5 distinguishes jobseekers according to whether they participate in transversal training and whether they list at least one shortage occupation among their preferred jobs. Six months after the intervention, treated jobseekers are 0.25 pp more likely to both participate in transversal training and search for a shortage occupation, an 11.7% increase relative to the control mean. However, the corresponding treatment effect for participating in transversal training without searching for a shortage occupation is 0.10 pp, and the difference between these two effects is not statistically significant ($p = 0.35$). The same pattern persists after 12 months. We therefore find no evidence that the increase in transversal training is disproportionately concentrated among jobseekers who also redirect their search toward shortage occupations.

At the same time, the intervention clearly reorients search toward shortage occupations, regardless of whether individuals decide to enroll in training. Among jobseekers who do not participate in transversal training, treatment increases the probability of searching for a shortage occupation by 0.73 pp after 6 months and 0.88 percentage points after 12 months, while reducing the probability of searching only for non-shortage occupations by 1.08 and 1.24 pp, respectively. The difference between these effects is statistically significant at both horizons. This suggests that jobseekers do respond to information about where labor demand is strongest. The absence of an effect on occupational training therefore does not appear to result from ignoring the shortage occupation information. Rather, the disconnect seems to arise at the human-capital investment margin, where jobseekers adjust the occupations they consider but do not commit to the specialized training required to enter them.

5.3 Effects on Employment

Finally, in Table A4 in the Appendix, we investigate to what extent the observed changes in training and job search behavior ultimately led to changes in employment probability. Given what we have shown so far, the intervention could have affected treated jobseekers' employment prospects through increased training participation, potentially after a "lock-in" period, a greater focus on shortage occupations in job search, or through a combination of both.

Table A4 shows no effect of the intervention on the probability of having experienced an employment spell 6 and 14 months after the intervention.²⁶ Focusing on the number of days employed or the probability of being employed 6 and 14 months after the intervention leads to similar conclusions. Although our data does not allow us to precisely distinguish employment effects by type of occupation (shortage versus non-shortage), proxies based on the industry of employment do not suggest that the intervention increased the probability that treated jobseekers enter shortage occupations.²⁷ In addition, Table A4 shows that our null employment effects still hold when we consider leaving unemployment as the outcome of interest, or when accounting only for relatively "stable" employment spells (i.e., those lasting more than five days). Finally, note that the absence of a negative effect on employment suggests that the increased participation in training induced by our intervention did not create significant lock-in effects for individuals who would otherwise have found a job.

Ultimately, this means that neither the increased participation in transversal training courses nor the greater focus on shortage occupations in job search were enough to generate a detectable increase in employment probability. One possible explanation is that, although following transversal trainings and targeting shortage occupations during job search can foster job finding, the changes in training and job search behaviors generated by the intervention were simply too modest to translate into substantial employment gains. Even under the optimistic assumption that training effects fully translated into employment, the

²⁶In an exploratory analysis, we examined if the increased (transversal) training participation and changes in job search strategy affected individuals' employment probability. Specifically, we estimated 2SLS regressions in which training participation and searching for a shortage occupation were instrumented by the treatment assignment. The estimates are too noisy to be interpreted with confidence, but they support our conclusion that the changes in training participation and job search induced by the intervention did not allow to significantly increase return into employment.

²⁷We checked whether treated jobseekers were more likely to find employment in a sector where there tends to be a higher share of occupations in shortage. This exercise relies on a number of assumptions, namely that jobs are spread evenly across occupations within a given sector. Since our employment data does not include information on occupation (only on the industry in which the employer is active), this is the closest we can get to exploring whether changes in job search behavior lead to more employment in shortage occupations.

implied increase in employment probability in our experimental population would amount to only 0.69%. Similarly, if all jobseekers who shifted their search toward shortage occupations successfully transitioned into such jobs, employment would have risen by just 1.67%. In the heterogeneity analysis, we show that the individuals most affected by the intervention had a low ex-ante probability of returning to employment. This supports the idea that the observed behavioral changes may not have been large enough to move them into employment.

Another explanation is that changes in training participation and job search are simply not effective to increase employment. For training, this hints to evidence on transversal programs being less effective than vocational courses in bringing the unemployed back to work (Stenberg and Westerlund, 2015; Woods et al., 2024), and occupational programs targeting high-demand occupations having a better record of fostering returns to employment (e.g., Katz et al., 2022; Grosz, 2020). For job search, this interpretation would suggest that searching a job in a shortage occupation is pointless if the jobseeker does not yet possess the necessary training or skills to fill those vacancies. This interpretation would indicate that, to generate meaningful employment gains, the information intervention would need to foster participation in trainings for shortage occupations rather than transversal trainings.

6 Robustness

Table 6 examines the sensitivity of our findings to alternative sample selection criteria and specifications. The first column reports our benchmark results for ease of comparison. Column (2) shows that our benchmark findings remain stable when we include control variables. Column (3) shows that our estimated effects are larger when we limit our sample exclusively to individuals who were still unemployed at the time that the email was sent. This suggests, as expected, that the effects are driven by those for whom the information was most relevant. Next, column (4) presents our estimated effects when including individuals who did not open the email in our sample. The estimated effects become somewhat attenuated when including these individuals since, by definition, non-openers could not have accessed the information and they thus introduce uninformative noise to our estimates. Finally, in column (5), we estimate Equation 1, but using only the sample of individuals who did not open the email, and thus where no one was treated. Reassuringly, we find no statistically significant effects in this placebo test.

7 Conclusion

In this paper, we conduct a large-scale field experiment in the labor market to examine how information about shortage occupations shapes unemployed jobseekers' training choices and occupational search. We contact the entire population of nearly 100,000 individuals who enrolled at the Walloon PES between July and December 2021, and send a randomly selected half of them information about shortage occupations and related training opportunities. We follow these individuals for up to 14 months after they receive the information and investigate whether the intervention affects training participation, the type of training chosen, and how these training responses relate to changes in job search behavior.

We show that the intervention increased training participation, with this response driven by transversal training rather than (shortage) occupational training. We provide evidence that this pattern is more likely driven by jobseekers' beliefs and preferences than by the supply of training opportunities. Jobseekers perceive transversal training as more useful for improving employment prospects, while transversal programs do not appear easier to access or more attractive in terms of duration, frequency, variety, or quality. Moreover, the intervention also reoriented job search toward shortage occupations, while the increase in transversal training was not disproportionately concentrated among jobseekers making this shift. This suggests that jobseekers respond to information about labor market tightness on two distinct margins, by redirecting their occupational search toward high-demand jobs and by investing in broader forms of skill development rather than occupation-specific training. These behavioral responses did not translate into detectable increases in employment.

Together, these findings contribute to the literature on workers' and jobseekers' training decisions and job search in high-demand occupations. While existing evidence shows that jobseekers can be encouraged to redirect their search toward occupations with better labor market prospects, less is known about how they translate shortage information into training choices when reskilling or upskilling is required. This distinction matters because occupational and transversal training are likely to have different returns, with demand-driven occupational programs often showing stronger employment effects.

Our results show that information on shortage occupations can increase training participation and shift job search toward occupations in demand, but that the training response is driven by transversal rather than occupational programs. This suggests that jobseekers may interpret shortage information as a signal to improve their skills while still avoiding narrower, occupation-specific investments. We therefore show that information can encourage

skill investment, but not necessarily in the type of training most directly linked to shortage occupations.

These findings have several policy implications. First, they suggest that broad information campaigns can generate interest in shortage occupations, as shown by the increase in shortage-oriented job search and by the fact that many treated jobseekers clicked on the information links. Yet the modest size of the effects also suggests that more targeted and personalized interventions may be needed to generate stronger responses. Evidence from job search recommendation tools indicates that tailored advice can be more effective than general information (Belot et al., 2019; Belot et al., 2026; Belot et al., 2025). When training is required to access high-demand occupations, however, recommendations alone may not suffice. Policies also need to account for jobseekers' occupational preferences and address their beliefs about the relative payoffs of transversal and occupational training.

More broadly, encouraging transitions toward shortage occupations likely requires going beyond information frictions. Redirecting job search toward high-demand occupations may be relatively low-cost for jobseekers, whereas enrolling in occupational training often requires a substantial investment of time and a stronger commitment to a new career path. Future policies should therefore pay closer attention to the features that make occupational training attractive, including perceived returns to training, working conditions in the target occupation, and training program characteristics such as format of delivery, duration and accessibility. Understanding how these factors interact with jobseekers' preferences is crucial if policymakers aim to increase participation in occupational training programs that can support transitions into high-demand jobs.

8 References

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Journal of Social Policy, 1-22.

9 Tables

Table 1: Summary Statistics

		Overall	Treated	Control	t-test
Gender	Woman	0.5218 (0.4995)	0.5229 (0.4995)	0.5206 (0.4996)	0.0023 (0.0032)
	Man	0.4782 (0.4995)	0.4771 (0.4995)	0.4794 (0.4996)	-0.0023 (0.0032)
Age	Age	32.6587 (11.2573)	32.6588 (11.2526)	32.6586 (11.2621)	0.0001 (0.0717)
Province	Hainaut	0.3844 (0.4865)	0.3838 (0.4863)	0.3851 (0.4866)	-0.0012 (0.0031)
	Liege	0.2809 (0.4494)	0.2813 (0.4496)	0.2805 (0.4492)	0.0008 (0.0029)
	Luxemburg	0.0823 (0.2749)	0.0819 (0.2743)	0.0827 (0.2755)	-0.0008 (0.0017)
	Namur	0.1489 (0.3560)	0.1494 (0.3564)	0.1485 (0.3556)	0.0008 (0.0023)
	Walloon Brabant	0.0961 (0.2947)	0.0967 (0.2956)	0.0955 (0.2939)	0.0013 (0.0019)
	Other	0.0073 (0.0854)	0.0069 (0.0828)	0.0078 (0.0879)	-0.0009* (0.0005)
Education	Max. Secondary	0.6014 (0.4896)	0.6016 (0.4896)	0.6011 (0.4897)	0.0005 (0.0031)
	Tertiary	0.2558 (0.4363)	0.2548 (0.4358)	0.2568 (0.4369)	-0.0020 (0.0028)
	Other	0.1428 (0.3499)	0.1436 (0.3507)	0.1420 (0.3491)	0.0016 (0.0022)
First-time Unemployed	No	0.7180 (0.4500)	0.7186 (0.4497)	0.7173 (0.4503)	0.0013 (0.0029)
	Yes	0.2820 (0.4500)	0.2814 (0.4497)	0.2827 (0.4503)	-0.0013 (0.0029)
Nationality	Belgian	0.8610 (0.3459)	0.8599 (0.3471)	0.8622 (0.3447)	-0.0023 (0.0022)
	Non-Belgian	0.1390 (0.3459)	0.1401 (0.3471)	0.1378 (0.3447)	0.0023 (0.0022)
# Obs		98,844	49,251	49,593	

Note: This table shows the pre-treatment characteristics of individuals included in the experiment, including any significant difference between the treatment and control group.

*** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

Table 2: Engagement with the Email

	(1) Overall Sample	(2) Treatment Group	(3) Control Group	(4) Difference
Opened email	0.5445 (0.4980)	0.5440 (0.4981)	0.5450 (0.4980)	-0.0010 (0.0032)
Clicked on at least one link	0.1261 (0.3319)	0.1392 (0.3461)	0.1131 (0.3167)	0.0261 *** (0.0028)
Number of clicks	0.1439 (0.4221)	0.1726 (0.4938)	0.1155 (0.3341)	0.0571 *** (0.0036)
Clicked on survey link	0.0945 (0.2925)	0.0781 (0.2683)	0.1108 (0.3139)	-0.0327 *** (0.0025)
Clicked on social network link	0.0026 (0.0510)	0.0019 (0.0440)	0.0033 (0.0571)	-0.0013 *** (0.0004)
Number of observations	98,844	49,251	49,593	

Note: This table shows descriptive statistics on email engagement for our overall sample, and the treatment and control groups separately. The last column shows the difference between treatment and control group means, with stars indicating the statistical significance of these differences. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

Table 3: Effects on Training Behavior

	(1) Training	(2) Transversal Training	(3) Occupational Training	(4) Short. Occ. Training	(5) Non-Short. Occ. Training
Short-term (6 months)					
β_1 (ITT)	0.0036* (0.0021)	0.0036** (0.0016)	-0.0001 (0.0015)	-0.0006 (0.0013)	0.0005 (0.0009)
Constant	0.0587*** (0.0015)	0.0358*** (0.0012)	0.0293*** (0.0010)	0.0222*** (0.0009)	0.0097*** (0.0006)
Long-term (12 months)					
β_1 (ITT)	0.0036 (0.0022)	0.0035* (0.0018)	-0.0002 (0.0016)	-0.0010 (0.0014)	0.0008 (0.0009)
Constant	0.0711*** (0.0016)	0.0443*** (0.0013)	0.0359*** (0.0011)	0.0276*** (0.0010)	0.0118*** (0.0007)
Observations	53,822	53,822	53,822	53,822	53,822

Note: This table shows the estimated effect of the intervention, using Equation 1, on training enrollment, 6 and 12 months after the intervention. “Training” covers enrollment in any training activity, and can be subdivided into “Transversal” or “Occupational” training. “Transversal Training” includes training activities that do not target a given occupation or sector, while “Occupational Training” includes programs focusing on the acquisition of skills specific to an occupation or sector. Finally, “Short. Occ. Training” encompasses occupational trainings that target shortage occupations, while “Non-Short. Occ. Training” includes occupational trainings that target occupations that are not in shortage. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

Table 4: Effects on Job Search

	(1) Search Short. Occ.	(2) Number Occ. Searched	(3) Connections to PES Website
Short-term (6 months)			
β_1 (ITT)	0.0099** (0.0042)	0.0221 (0.0189)	-0.1268** (0.0564)
Constant	0.5881*** (0.0030)	3.2871*** (0.0133)	1.6747*** (0.0398)
Long-term (14 months)			
β_1 (ITT)	0.0114*** (0.0042)	0.0314 (0.0208)	-0.1573 (0.1331)
Constant	0.6120*** (0.0030)	3.6304*** (0.0146)	5.4415*** (0.0939)
Observations	53,822	53,822	53,822

Note: This table shows the estimated effect of the intervention, using Equation 1, on employment and job search behavior. “Search Short. Occ.” is defined as listing at least one shortage occupation in one’s occupational preferences. “Number Occ. Searched” is the total number of occupations listed as occupational preferences. “Connections to PES Website” is the number of times a jobseekers logs into their personal space on the PES website. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

Table 5: Joint Responses in Transversal Training and Search for Shortage Occupations

	(1) Training & No Search SO	(2) No Training & Search SO	(3) Training & Search SO	(4) No Training & No Search SO
Short-term (6 months)				
β_1 (ITT)	0.0010 (0.0010)	0.0073* (0.0043)	0.0025** (0.0013)	-0.0108** (0.0042)
Constant	0.0140*** (0.0007)	0.5661*** (0.0030)	0.0218*** (0.0009)	0.3977*** (0.0030)
Long-term (12 months)				
β_1 (ITT)	0.0010 (0.0011)	0.0088** (0.0042)	0.0025* (0.0015)	-0.0124*** (0.0042)
Constant	0.0150*** (0.0008)	0.5826*** (0.0030)	0.0293*** (0.0010)	0.3728*** (0.0029)
Observations	53,822	53,822	53,822	53,822

Note: This table shows the estimated effect of the intervention, using Equation 1, on the combination of training and job search behavior, 6 and 12 months after the intervention. “Training & No Search SO” is a dummy equal to one when an individual enrolls in transversal training, but does not list any shortage occupation in their preferred occupations (i.e., they only search for occupations that are not in shortage). Conversely, “No Training & Search SO” is a dummy equal to one when an individual lists at least one shortage occupation in their preferred occupations, but does not enroll in any transversal training. “Training & Search SO” is equal to one for individuals who both enroll in transversal training and list at least one shortage occupation in their preferred occupations. “No Training & No Search SO” is equal to one when individuals neither enroll in transversal training nor list any shortage occupation in their preferred occupations. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

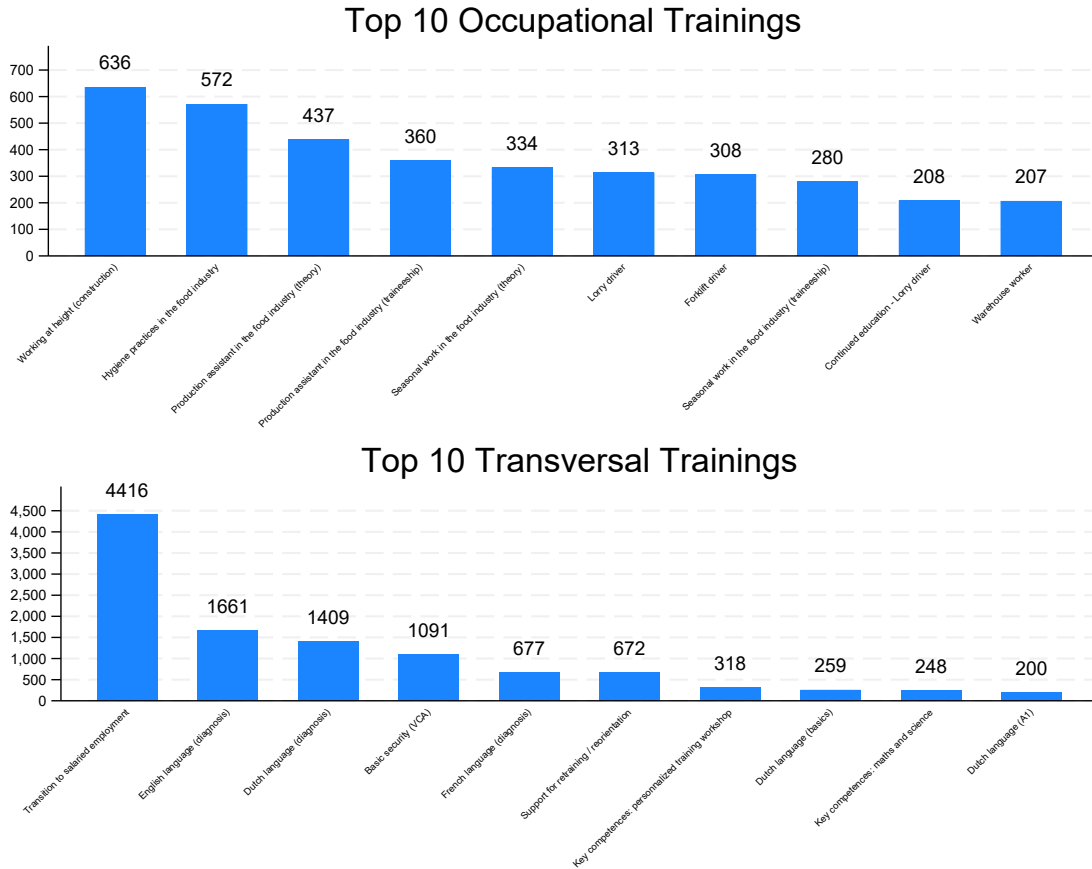
Table 6: Robustness Checks

	(1) Benchmark	(2) With Controls	(3) Unemployed in $t = 0$	(4) Keeping Non-Openers	(5) Placebo Non-Openers
Short-term estimates					
Any Training	0.0036* (0.0021)	0.0038* (0.0021)	0.0051* (0.0029)	0.0018 (0.0014)	-0.0003 (0.0019)
Transversal Training	0.0036** (0.0016)	0.0036** (0.0016)	0.0048** (0.0024)	0.0027 (0.0011)	-0.0010 (0.0014)
Occupational Training	-0.0001 (0.0015)	0.0001 (0.0015)	-0.0003 (0.0020)	-0.0003 (0.0010)	-0.0005 (0.0013)
Short. Occ. Training	-0.0006 (0.0013)	-0.0004 (0.0013)	-0.0019 (0.0018)	-0.0008 (0.0009)	-0.0009 (0.0012)
Non-Short. Occ. Training	0.0005 (0.0009)	0.0005 (0.0009)	0.0007 (0.0012)	0.0005 (0.0006)	0.0005 (0.0007)
Search Short. Occ.	0.0099** (0.0042)	0.0108*** (0.0039)	0.0090* (0.0054)	0.0065** (0.0031)	0.0023 (0.0046)
Number Occ. Searched	0.0221 (0.0189)	0.0269 (0.0172)	0.0284 (0.0241)	-0.0025 (0.0139)	-0.0320 (0.0207)
Connections to PES Website	-0.1268** (0.0564)	-0.1227** (0.0564)	-0.1199 (0.0801)	-0.0499 (0.0363)	0.0432 (0.0423)
Long-term estimates					
Any Training	0.0036 (0.0022)	0.0042* (0.0022)	0.0058* (0.0032)	0.0013 (0.0015)	-0.0013 (0.0021)
Transversal Training	0.0035* (0.0018)	0.0038** (0.0018)	0.0055** (0.0026)	0.0011 (0.0012)	-0.0017 (0.0016)
Occupational Training	-0.0002 (0.0016)	0.0003 (0.0016)	-0.0004 (0.0022)	-0.0005 (0.0011)	-0.0009 (0.0015)
Short. Occ. Training	-0.0010 (0.0014)	-0.0006 (0.0014)	-0.0021 (0.0020)	-0.0011 (0.0010)	-0.0012 (0.0013)
Non-Short. Occ. Training	0.0008 (0.0009)	0.0009 (0.0009)	0.0009 (0.0013)	0.0006 (0.0006)	0.0005 (0.0009)
Search Short. Occ.	0.0114*** (0.0042)	0.0121*** (0.0038)	0.0120** (0.0054)	0.0076** (0.0031)	0.0031 (0.0045)
Number Occ. Searched	0.0314 (0.0208)	0.0348* (0.0192)	0.0332 (0.0265)	0.0006 (0.0155)	-0.0362 (0.0231)
Connections to PES Website	-0.1573 (0.1331)	-0.1689 (0.1327)	-0.2646 (0.1900)	-0.0703 (0.0848)	0.0374 (0.0960)
Observations	53,822	53,709	32,464	98,844	45,022

Note: This table shows the sensitivity of our benchmark estimates to: the inclusion of control variables in Column (2), the exclusion of individuals who were not unemployed anymore at the time of receiving the treatment in Column (3), and the inclusion of individuals who did not open the email in Column (4). Column (5) shows the results of a placebo test using only individuals who did not open the email. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

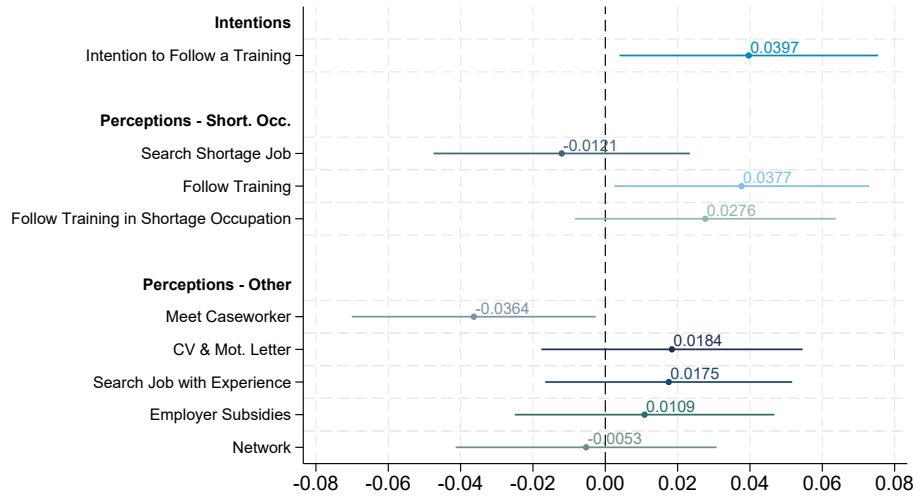
10 Figures

Figure 1: Top 10 Trainings



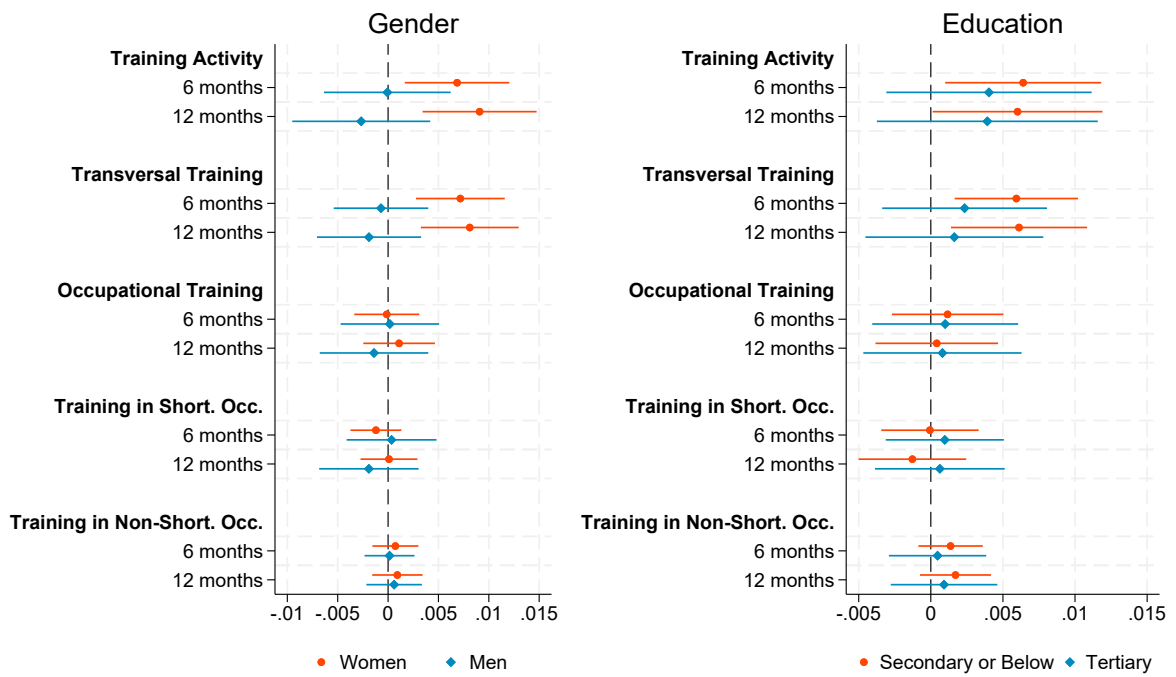
Notes: This figure shows the ten training names that occur most frequently in our data on training participation (pre- and post-intervention), for occupational trainings (total instances=123,812) and transversal trainings (total instances=112,857). Each bar represents the frequency of appearance of that title in the data.

Figure 2: Effects on Beliefs and Intentions



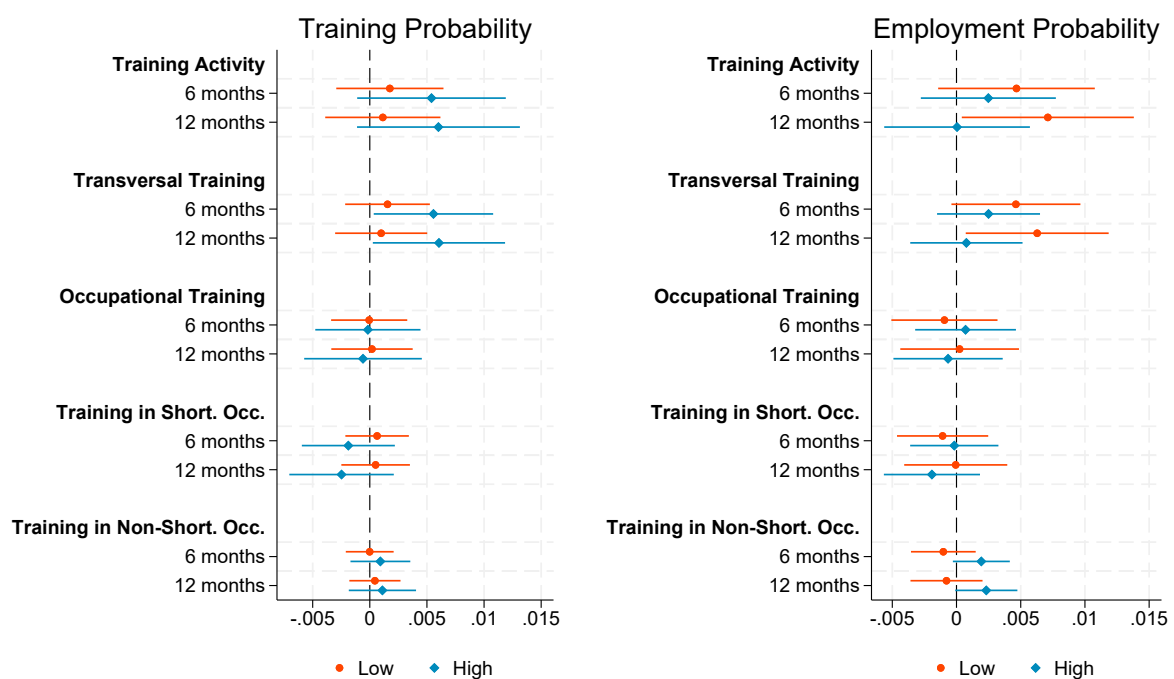
Notes: This figure shows the estimated effect, using Equation 1, of being sent the information treatment on training intentions and perceptions about the usefulness of various job search strategies. “Intentions” show treatment effects on the probability of answering “yes” when asked about having the intention to enroll in a training program in the coming year (control mean=0.55). “Perceptions – Shortage Occupations” refer to the probability of considering job search strategies related to shortage occupations as having a high or very high impact on job finding probability. Strategies in this group include searching for a job in a shortage occupation (control mean=0.41), following a training (control mean=0.59), and following a training in a shortage occupation (control mean=0.48). “Perceptions – Other Strategies” refer to the probability of considering job search strategies unrelated to shortage occupations as having a high or very high impact on job finding probability. Strategies in this group include meeting your caseworker (control mean=0.38), having a good CV and motivation letter (control mean=0.49), searching for a job in which one has experience (control mean=0.66), using employer subsidies (control mean=0.44), and using one’s network (control mean=0.52). Confidence intervals are shown at 95%.

Figure 3: Heterogeneous Effects - Personal characteristics



Notes: This figure shows the estimated effects of the intervention, using Equation 1, by gender (left panel) and education level (right panel). Confidence intervals are shown at 95%.

Figure 4: Heterogeneous Effects - Ex-ante Training and Employment Probability



Notes: This figure shows the estimated effects of the intervention, using Equation 1, by ex-ante probability of enrolling in training (left panel) and finding any employment (right panel) within 6 months of the intervention. Confidence intervals are shown at 95%.

Appendix

A.1 Shortage Occupations in Wallonia

Every year, the Walloon PES identifies the list of occupations that are currently in shortage in the region. This list is made using two types of tools: statistical evidence from employer surveys and qualitative expert knowledge. The first tool consists in a yearly survey in which employers are asked about whether or not (and how quickly) they were able to fill job openings for different types of occupations. The PES considers occupations to be in shortage if (i) employer satisfaction with the job filling for that occupation is below the median, and (ii) the average time to fill the opening is above the median. The list that is made on the basis of the employer survey is then submitted to the opinion of experts who work within the PES or in training centres. Experts can then add or remove occupations from the list based on their knowledge of the field.

The PES also makes a distinction between “critical occupations”, defined as those for which employers face strong hiring difficulties, and “shortage occupations”, defined as critical occupations for which there is a quantitative shortage of candidates with the appropriate qualifications. A quantitative shortage is defined as there being less than 15 unemployed jobseekers with the appropriate qualification for every ten job openings in an occupation. In the paper, we grouped “critical occupations” and “shortage occupations” together, and simply referred to them as shortage occupations.

The following occupations were considered in shortage when our experiment took place:

- Accountant
- Accounting Expert
- Agricultural and Technical Mechanic
- Air Conditioning/Ventilation Fitter
- Aircraft Maintainer
- Architect
- Army Officer
- Automation Technician
- Automotive Maintenance and Diagnostic Technician (MDA)
- Baker
- Barman
- Building Maintenance Worker
- Bus Driver

- Business Analyst
- Business Engineer
- Butcher
- Car Repairer
- Caregiver
- Carpenter
- Cementer
- Chef/Cook
- Civil Engineering Mechanic
- Coach Driver
- Commercial Agent (Real Estate Sector)
- Construction and Road Maintenance Equipment Operator
- Construction Draftsman
- Construction Supervisor/Manager
- Construction Worker
- Customs Agent
- Data Scientist / Analyst
- Dispatcher in Transport and Logistics
- Domestic Helper
- Driving Instructor
- Earthmoving Equipment/Machine Operator
- Electrical Installer/Maintenance Technician
- Electromechanic
- Electronics Maintenance Technician
- Energy Consultant
- Facilities Manager in Food Industry
- Fast-food Restaurant Manager
- Financial/Credit Analyst/Advisor
- Florist
- Gardener
- General Practitioner (GP)
- General/Specialist Nurse
- Glassmaker
- Hairdresser
- Healthcare Assistant
- Heating Maintenance and Operation Technician

- Industrial and Logistics Manager
- Industrial Cleaner
- Industrial Installation and Maintenance Technician/Manager
- Industry Planning Officer
- Information Systems Administrator
- Insulation Worker
- Insurance Advisor
- IT Analyst
- IT Developer
- IT Project Manager
- Kitchen Installer
- Laboratory Control Technician
- Logistics Operations / Warehouse Manager
- Logopedist
- Machines Technician (Metal)
- Manufacturing Operator in the Chemical Industry
- Mason
- Mechanical Fitter
- Medical Imaging Technologist
- Metal Worker
- Motion Designer
- Mover
- Multi-skilled/Truck mechanic
- (Industrial) Painter
- Partitions/False Ceilings Fitter
- Pastry Chef
- Physics or Chemical Transformation Equipment Operator
- (Industrial) Pipe Fitter
- Plasterer
- Plumbing and Heating Fitter
- Police Inspector
- Prevention Advisor
- Production line operator (food industry)
- Production Manager/Technician
- Production/Quantity Planner
- Property/Building Manager

- Quality and Regulations Manager
- R&D Manager
- (Hotel) Receptionist
- Refrigeration Technician
- Rejoiner/Brick Sealer
- Research Laboratory Technician
- Retail Store Manager
- Road Worker
- Roof Builder
- Sales Assistant
- Sales Manager
- Sales Representative for consumer goods
- Sales Representative for professional equipment goods
- Sales Representative in business services
- Scaffolder
- Security Guard
- Security System Installer
- Slaughterer / Meat Processing Worker
- Sports Instructor
- Swimming Pool Installer
- Technical Compliance Officer
- Technical Sales Representative
- Technician in the Performing Arts and Events
- Textile Manufacturing Machine Operator
- Tiler
- Tourism Product Manager
- Tower Crane Operator
- Travel Agent
- Truck Driver
- Valet / Chambermaid / Housekeeper
- (Head) Waiter
- Warehouse Worker/Manager
- Waste Sorter
- Waterproofing Contractor
- Web Developer
- Welder

A.2 Conditions for Job Search Exemptions While in Training

At the time of our experiment, jobseekers were exempted from their job search obligations when they followed a training. To receive this exemption, they had to fulfill the following obligations:

1. Be a UI recipient living in Wallonia.
2. Have finished their last studies (preceding the first receipt of UI benefits) for at least 2 years.
3. Have been unemployed for at least 312 days over the past two years, except if the training considered is for a shortage occupation.
4. Not have a higher education (post-secondary) degree, except if this degree offers few opportunities on the labor market.
5. Not have already benefited from an exemption to pursue a training (except for prolongation requests).

On top of the fact that jobseekers are exempted from their job search obligations during the duration of their training and therefore keep the right to receive UI benefits, they are also entitled to a travel expense allowance and extra childcare allowances. Moreover, if they complete a training for a shortage occupation, they are offered a €350-2000 bonus and receive special coaching to prepare them for interviews.

During the period of the experiment, unemployment benefits in Belgium were largely unlimited in time. Therefore, enrolling in training is not a potential way of extending benefit duration in our context.

A.3 Treatment and Control Emails

Figure A1: Control Group Email



Figure A2: Treatment Group Email

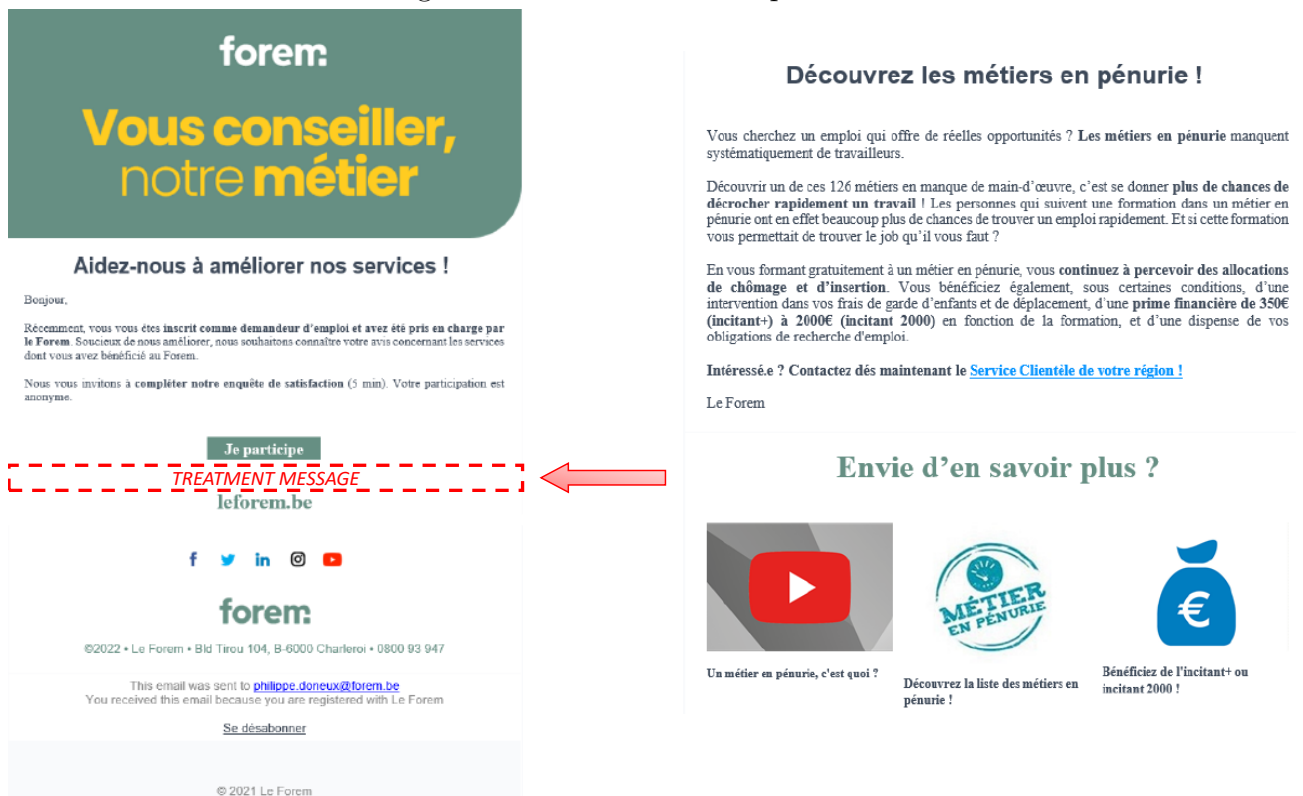


Table A1: Engagement with the Treatment Information

Clicked on “Client Service” link	0.0147 (0.1204)
Clicked on video	0.0165 (0.1274)
Clicked on list of shortage occupations	0.0466 (0.2108)
Clicked on information about financial incentive	0.0124 (0.1108)
Clicked on at least one treatment link	0.0765 (0.2659)
Number of observations	49,251

Note: This table shows descriptive statistics on email engagement among the treatment groups. Statistics on clicking behavior are shown for the sample of email openers. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

A.4 Survey on Training Intentions and Beliefs

A.4.1 Survey Questions

The intervention took place in the context of a satisfaction survey which was sent to recently unemployed jobseekers. The invitation to complete the survey was sent to both the treated and the control groups, meaning we were able to add several questions to the survey in order to measure effects on beliefs and intentions about training and job search. A challenge here was to obtain information on perceptions about shortage occupations and related trainings from both the treated and control groups, while avoiding as much as possible to inadvertently prime the control group toward thinking about shortage occupations. With this in mind, we attempted to dilute questions on shortage occupations with questions on unrelated topics.

Specifically, we added three questions to the survey. First, we asked jobseekers whether they had the intention of enrolling in any training in the coming year, to which they could answer yes or no. This question was intended to determine to what extent the treatment email had affected the training intentions of treated jobseekers; we expect treated jobseekers to be relatively more interested in training than those not treated. We purposefully did not ask about training for shortage occupations to avoid inadvertently treating individuals from the control group, who would otherwise have been primed to think about shortage occupations. We coded this outcome as a dummy variable equal to one if the individual answered that they had the intention of following a training, and zero otherwise.

Second, we wanted to determine whether the intervention had affected perceptions (or beliefs) about how useful it is to focus one's job search strategy on shortage occupations and (related) trainings. We asked jobseekers how big of an impact they thought a given search strategy would have on their chances of finding a job. They could choose between "no impact", "little impact", "average impact", "high impact", and "very high impact". We asked them to evaluate the following job search strategies: (i) frequently meeting with one's caseworkers, (ii) following a coaching session to have a good CV and motivation letter, (iii) focusing one's job search on shortage occupations, (iv) using employer subsidies, (v) focusing one's job search on occupations in which they have experience, (vi) following trainings, (vii) exploiting one's professional network, and (viii) following a training in a shortage occupation. The options we expected to be affected by our information treatment were options (iii), (vi), and (viii), i.e., those relating to shortage occupations and/or training activities. When building these outcomes variables, we coded dummy variables equal to one if the respondent considered a given strategy to have a "high impact" or "very high impact" on their chances of finding a job, and zero if they answered it had "no impact", "little impact" or an "average impact".

Third, we asked jobseekers to mark their degree of agreement with a series of statements on how supported by the PES they felt. Specifically, we asked them to state whether they “completely disagreed”, “disagreed”, “had mixed views”, “agreed” or “agreed strongly” with four statements. The statements related to the support offered to jobseekers by the PES in terms of (i) number of training opportunities offered, (ii) financial help when seeking training, (iii) waiver of job search requirements when enrolled in training, and (iv) number of job openings sent to them.

Finally, we also asked respondents from the treatment group whether they had read the information on shortage occupations in the email. Our survey showed that, among those who answer the survey, only 56.7% of individuals assigned to the treatment group had actually read the treatment information before answering the survey. This could be due to the fact that the survey link was located above the treatment information in the email; respondents might have therefore clicked on the survey before looking further in the content of the email.

A.4.2 Selection into the Satisfaction Survey

Table A2 shows the characteristics of individuals who opened the email (column 2) and clicked on the survey (column 3) compared to the overall sample (column 1). First, it shows that women are more likely to open the email than men and, conditional on opening, even more likely to answer the survey. Second, older people are slightly more likely to open the email, but much more likely to answer the survey than younger individuals, conditional on opening. Third, people with a higher level of education (tertiary or university degree) are slightly more likely to open the email than individuals with a lower level of education (secondary degree or less). Those most likely to open and, especially, click on the survey are those with an “Other” degree, i.e., those with an apprentice degree or a diploma from outside the EU. Fourth, non-Belgians appear to be overrepresented among email openers, and even more so among survey clickers. Fifth, individuals who had already been unemployed in the past are slightly less likely to open the email but, conditional on opening, they are relatively more likely to answer the survey than the first-time unemployed. Finally, there does not seem to be any selection of email openers and survey clickers on the basis of where individuals live. Overall, we can thus conclude that women, non-Belgians, and older individuals are generally overrepresented among those who open the email and click on the survey compared to the overall sample.

In total, 3,049 individuals answered the survey, of which 1,252 (41%) belonged to the treatment group and 1,797 (59%) belonged to the control group. In other words, 2.54% of

individuals in the treatment group completed the survey, against 3.62% of the control group. Although the sample of individuals who completed the survey is a selected sub-sample of our overall target population, treated and control individuals do not appear to self-select differently, based on observable variables, into answering the survey.

Table A2: Summary Statistics – Email Openers and Survey Clickers

		Overall Sample	Opened Email	Clicked on survey
Gender	Woman	0.5218	0.5403	0.5791
		(0.4995)	(0.4984)	(0.4938)
	Man	0.4782	0.4597	0.4209
		(0.4995)	(0.4984)	(0.4938)
Age	Age	32.6587	33.0565	37.9538
		(11.2573)	(11.5651)	(12.7970)
Province	Hainaut	0.3844	0.3739	0.3808
		(0.4865)	(0.4838)	(0.4856)
	Liege	0.2809	0.2851	0.2854
		(0.4494)	(0.4515)	(0.4516)
	Luxemburg	0.0823	0.0828	0.0786
		(0.2749)	(0.2756)	(0.2692)
	Namur	0.1489	0.1485	0.1463
		(0.3560)	(0.3556)	(0.3534)
Walloon Brabant	0.0961	0.1016	0.1022	
	(0.2947)	(0.3021)	(0.3029)	
Other	0.0073	0.0081	0.0068	
	(0.0854)	(0.0894)	(0.0820)	
Education	Max. Secondary	0.6014	0.5658	0.5182
		(0.4896)	(0.4957)	(0.4997)
	Tertiary	0.2558	0.2749	0.2433
		(0.4363)	(0.4465)	(0.4291)
Other	0.1428	0.1593	0.2385	
	(0.3499)	(0.3659)	(0.4262)	
First-time Unemployed	No	0.7180	0.7003	0.7494
		(0.4500)	(0.4581)	(0.4334)
	Yes	0.2820	0.2997	0.2506
		(0.4500)	(0.4581)	(0.4334)
Nationality	Belgian	0.8610	0.8386	0.7619
		(0.3459)	(0.3679)	(0.4259)
	Non-Belgian	0.1390	0.1614	0.2381
		(0.3459)	(0.3679)	(0.4259)
# Obs		98,844	53,822	5,179

A.5 Additional Findings

Table A3: Additional Training Outcomes

	(1) Days in Training	(2) Training > 5 Days	(3) Training > 30 Days	(4) Complete Training	(5) Pre-Training
Short-term (6 months)					
β_1 (ITT)	0.2513 (0.2271)	0.0035* (0.0018)	0.0029* (0.0016)	0.0036** (0.0017)	0.0012 (0.0020)
Constant	6.2356*** (0.1603)	0.0459*** (0.0013)	0.0354*** (0.0011)	0.0405*** (0.0012)	0.0554*** (0.0014)
Long-term (12 months)					
β_1 (ITT)	0.2437 (0.3419)	0.0040** (0.0020)	0.0027 (0.0018)	0.0033* (0.0019)	-0.0008 (0.0026)
Constant	9.6106*** (0.2413)	0.0561*** (0.0014)	0.0444*** (0.0013)	0.0496*** (0.0013)	0.0976*** (0.0018)
Observations	53,822	53,822	53,822	53,822	53,822

Note: This table shows the estimated effect of the intervention, using Equation 1, on training duration and completion, 6 and 12 months after the intervention. “Days in Training” is the total number of days individuals are registered as being enrolled in a training. “Training > 5 Days” is a dummy equal to one if the individual has been registered in a training lasting more than 5 days (i.e., more than a work week). “Training > 30 Days” covers training activities that last more than 30 days (i.e., more than a month). “Complete Training” is a dummy equal to one if the individual has participated and completed any training activity (i.e., “Any Training” where activities are recoded to 0 if they are not completed). “Pre-Training” includes any activity that takes place before an actual training, such as information sessions, selection tests, or screenings. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

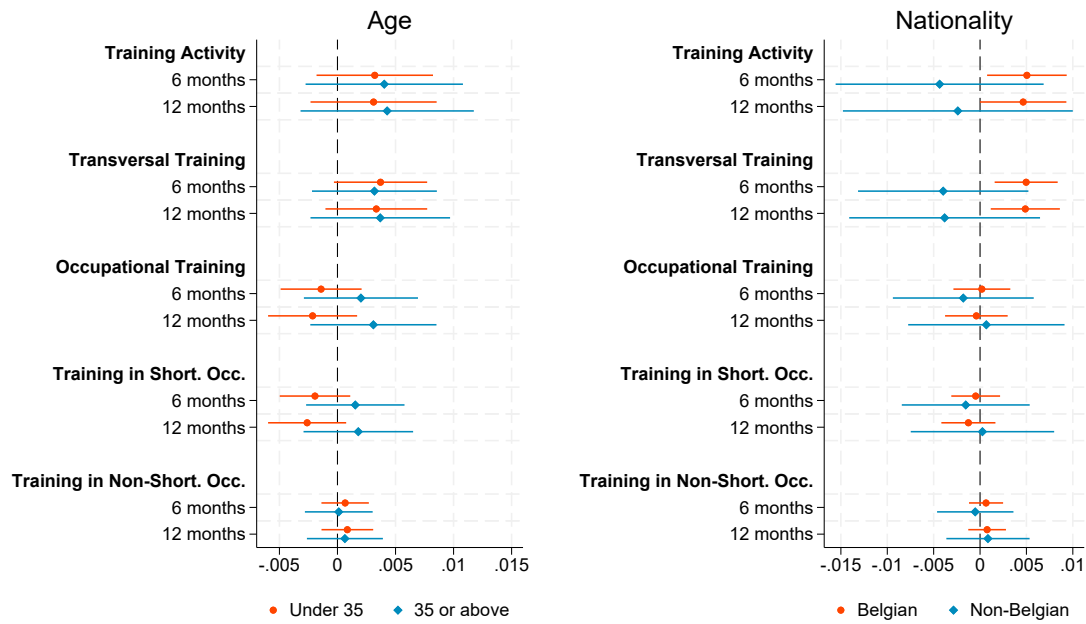
Table A4: Effects on Employment Outcomes

	(1) Ever Leave UI	(2) Leave UI (> 5 days)	(3) Any Employment	(4) Employment (> 5 days)
Short-term estimates				
β_1 (ITT)	0.0018 (0.0042)	0.0012 (0.0043)	0.0024 (0.0043)	-0.0014 (0.0042)
Constant	0.6054*** (0.0030)	0.5631*** (0.0030)	0.5161*** (0.0030)	0.3902*** (0.0030)
Long-term estimates				
β_1 (ITT)	-0.0010 (0.0034)	-0.0005 (0.0037)	-0.0009 (0.0039)	-0.0023 (0.0042)
Constant	0.8039*** (0.0024)	0.7604*** (0.0026)	0.7049*** (0.0028)	0.6193*** (0.0030)
Observations	53,822	53,822	53,822	53,822

Note: This table shows the sensitivity of our benchmark estimates to using alternative definition for our employment outcome. “Ever Leave UI” and “Leave UI (> 5 days)” are dummies equal to one if the individual is ever observed leaving unemployment (for at least five days in the latter case). “Any Employment” is defined as having experienced at least one employment spell since the intervention. “Employment (> 5 days)” considers only employment spells that lasted more than five days. *** $p < 0.01$ ** $p < 0.05$ * $p < 0.1$

A.6 Additional Heterogeneity Analyses

Figure A3: Heterogeneous Effects - Other Personal Characteristics



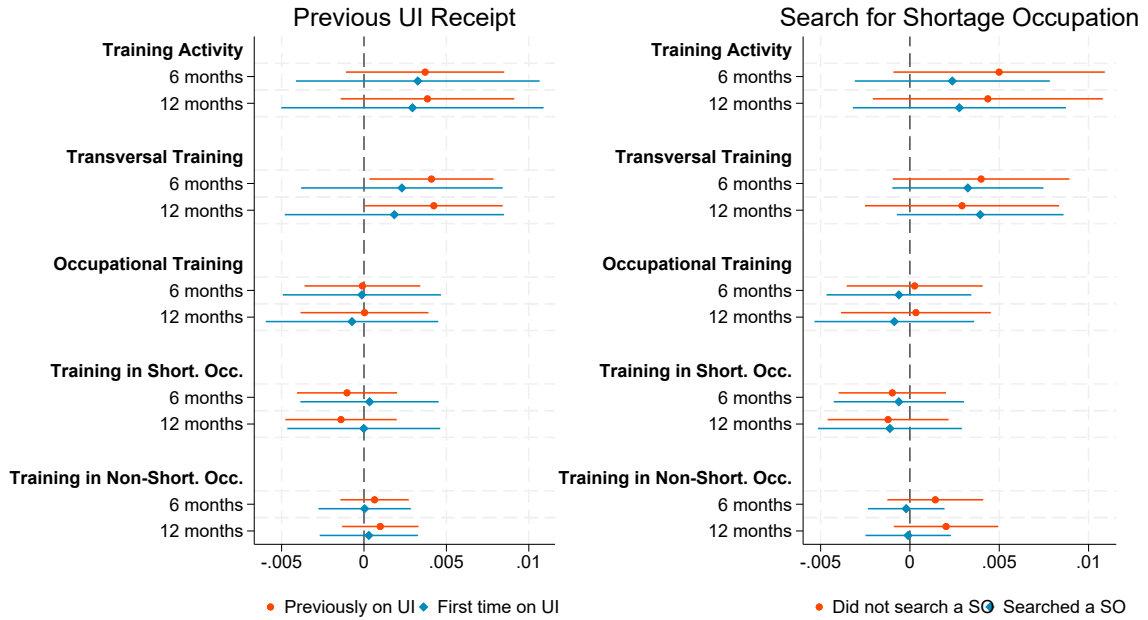
Notes: This figure shows the estimated effects of the intervention, using Equation 1, by age (left panel) and nationality (right panel). Confidence intervals are shown at 95%.

Figure A4: Heterogeneous Effects - Unemployment Status



Notes: This figure shows the estimated effects of the intervention, using Equation 1, by unemployment (left panel) and benefit receipt (right panel) status in $t = 0$. Confidence intervals are shown at 95%.

Figure A5: Heterogeneous Effects - Type of Unemployed



Notes: This figure shows the estimated effects of the intervention, using Equation 1, based on whether the individual was unemployed for the first time (left panel) and whether they listed a shortage occupation in their preferred occupations at the start of the intervention. Confidence intervals are shown at 95%.

A.7 Descriptive Statistics on Trainings

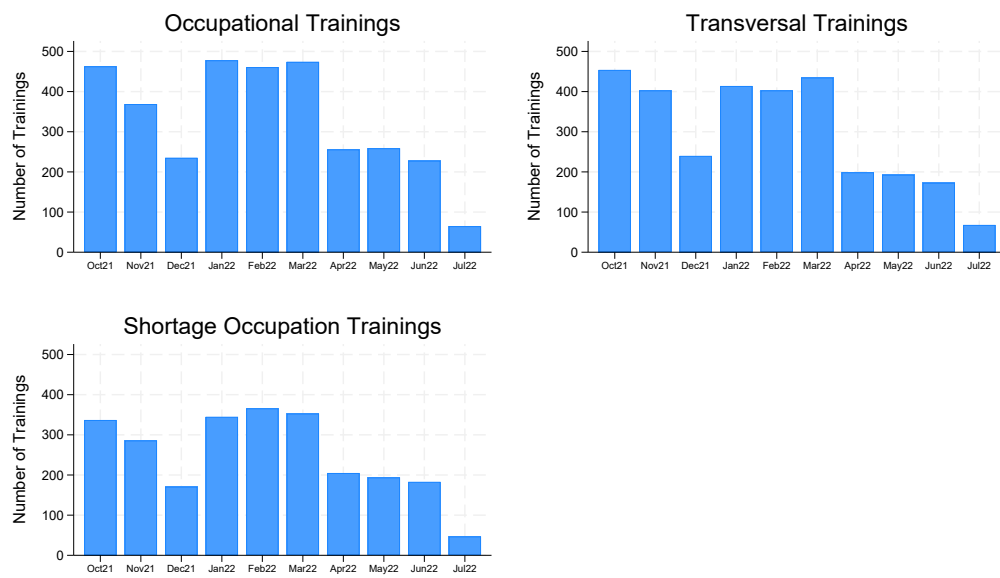
Figure A6 and Figure A7 show descriptive statistics, at the training level, on training activities observed in our data. This training data contains rich information, but comes with the caveat that we cannot rule out that the subset of observed trainings is not biased by the intervention itself. Since the intervention increased participation in transversal trainings, the number of observed transversal trainings compared with occupational trainings might be higher than it would have been without the intervention.

Figure A6: Distribution of Training Duration, by Training Type



Notes: This figure shows the distribution of training duration, for all training activities of a given type (Transversal, Occupational, and Shortage Occupation).

Figure A7: Number of Distinct Training Activities, by Training Type (Oct 2021-July 2022)



Notes: This figure shows the average number of trainings with distinct names that started each month over the course of the intervention, by training type (Transversal, Occupational, and Shortage Occupation).